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easyspec News

easyspec 1.3.0.0: now also for imaging photometry

The new version of easyspec improves the extraction of spectra and goes beyond long-slit spectroscopy: the new module `photometry` takes you from the raw images of a field to calibrated magnitudes, light curves and surface-brightness profiles, with the same simple and explicit style of the spectroscopy modules.

What is new in spectroscopy?

- **Optimal extraction**: the spectra can now be extracted with the optimal-extraction algorithm of Horne (1986, PASP 98, 609), which weights each pixel by the spatial profile of the source and its noise, and rejects cosmic rays and bad pixels: a higher S/N than the simple sum, especially for faint targets.
- **Automatic identification of the arc lines**: built-in line lists of the most common lamps (HgI, NeI, ArI, HeI, CuI, CdI, ZnI, KrI, XeI, FeI and ThAr, from PypeIt) for the wavelength calibration.

What is new in photometry?

- **Aperture and PSF photometry**: an empirical PSF for crowded fields such as globular clusters, and quality flags for every star.
- **Photometric calibration**: zero points, extinction and color terms from standard stars, with the Landolt and Clem & Landolt catalogues built in.
- **Color-magnitude diagrams**, in calibrated magnitudes and colors.
- **Surface photometry**: brightness profiles of clusters and galaxies, with King and Sérsic fits.
- **Exoplanet transits**: differential photometry, light curves and the fit of a transit model.
- **Color images**: `cleaning.generate_colored_image()` combines any number of bands with any colors, with many options to make the image beautiful. It can even remove the stars, leaving only the nebula or galaxy behind them.
- **WCS preserved**: trimming, alignment and stacking now keep the astrometry of the images.



The Crab Nebula from raw VLT/FORS1 images in B, V and R, with the stars removed by easyspec.

Four new tutorials

Each tutorial starts from raw data of large telescopes (ESO/WFI, NTT/EFOSC2 and VLT/FORS1), included in the repository:

- **color images of a nebula**: the Crab Nebula, from the raw frames to the image above;
- **aperture photometry of an open cluster**: the Jewel Box (NGC 4755), from the reduction to the calibrated color-magnitude diagrams;
- **PSF photometry of a globular cluster**: M79 (NGC 1904), with its color-magnitude diagram and King profile;
- **an exoplanet transit**: differential photometry and the fit of the transit.

The tutorials and their data are available **on GitHub**:

`https://github.com/ranieremenezes/easyspec/tree/main/tutorial`

and the documentation of all functions at `https://easyspec.readthedocs.io`.